

THERMODYNAMICS

ChE 322 – FALL 2024

Unique # - 15505

T, Thu from 11:00 a.m. – 12:15 p.m. (CPE 2.216)

Recitations: W 1:00 – 3:00 PM (CPE 2.216)

Instructor - Prof. Venkat Ganesan

TEXTBOOK: (Main - Required) *Engineering & Chemical Thermodynamics*, 2nd ed., Milo D. Koretsky, Wiley; (Alternate - Optional) J. M. Smith, H. C. Van Ness, and M. M. Abbott, *Introduction to Chemical Engineering Thermodynamics* (8th edition), McGraw-Hill (2004).

PREREQUISITES: CHE 210; CHE 317; M 427K; CH 353 (all with C- or above).

RECITATION SESSIONS (MANDATORY): Wednesday 1:00 - 3:00 p.m.

OFFICE HOURS: Monday 1:30 – 3:30 pm in CPE 3.414.

TEACHING ASSISTANT: TJ Duncan (Office Hrs @ CPE 2.222: Monday – 4:00 – 5:30 pm; Tuesday – 4:00 – 5:30pm); (tj_duncan@utexas.edu)

TUTORS: Sankalpa Bhuma (Office hours @CPE 2.222: Tuesday 6 – 8 pm) (sankalpa@utexas.edu)

HOMEWORK POLICY: There will be a homework assignment for nearly every week during the semester. *Homeworks will not be collected, however, you will have a closed book/closed notes quiz over the homework set that is scheduled for that day during the recitation time.* These recitation quizzes will be composed of problems very similar to homework problems (with some simple modifications) and/or verbatim homework problems. It may also include one or two concept questions similar to the ones we will discuss in lectures. The recitation quiz grades will contribute 25% to your final grade. **Recitation quizzes cannot be made up, however your two lowest recitation quiz scores will be dropped.** If more than two recitation quiz is missed, the additional missed quizzes will be scored as a zero.

You may work together with your classmates on homework but don't deny yourself the opportunity to "struggle" with difficult concepts and problems (by having friends "spoon-feed" you). It is my opinion that one learns a great deal in this endeavor....homework problems are your means of "practicing" engineering just as an athlete practices to improve his/her athletic ability.

Scholastic dishonesty will not be tolerated and incidents of dishonesty will be reported to the Dean of Student Affairs.

LECTURE ATTENDANCE: It is important and expected that you will attend all lectures. I will try to do my part to make the lectures interesting and worth attending. There is a strong correlation between class attendance and participation in course

activities and understanding and performance in the course, therefore, I strongly advise you to attend lectures.

To ensure that there is a consistent pattern of attendance, there will be a number (almost every lecture) of very short lecture quizzes. These will be held in the beginning of the lecture and will typically not require any preparation and take very little time (< 5 mins). Sometimes, these will be based on a problem, I may post as “pre-lecture” problem. These quizzes will only be graded on 1 or 0 scale. You will be allowed to drop the two lowest quiz scores.

Towards the later portion of the course, you will need a scientific calculator which can solve simultaneous nonlinear equations (TI-Nspire).

COVID-19 ACCOMODATIONS: We strongly recommend that vaccinated and unvaccinated students wear masks inside the classroom. If you contract COVID or do not feel well and suspect that you may have COVID, please do not come to lecture or discussion section.

The lectures will not be recorded. If you are sick, and have to miss a lecture, please contact me in advance so that we can identify some solutions.

Understandably, people may continue to fall sick and may have to miss recitation quizzes and mid-terms. As you will see below, we have set up a number of means to make up for missed recitation quizzes and mid-term exams. However, considering the size of class, it will not be possible to accord additional “make-up” recitation quizzes and mid-term exams. So, I would urge you to do well in every one of the quizzes you attend and not necessarily bank on being able to drop some of them.

EXAMS:

Midterm and final exams will be very challenging and you will need to study extensively to perform well. These exams will be closed book and closed notes. **You will be allowed one "formula card" made up of a 3 in. x 5 in. card** with anything that you like written on it (both sides may be used). **Exams cannot be made up.....if a Midterm Exam must be missed due to illness or family crisis then the Final Exam score will be substituted for the missed exam.** This can only be done once and students that must miss more than one Midterm Exam should see the Deans Office (ECJ 2.200) regarding dropping the course. If you have questions about the manner in which your exam was graded then write a note describing the discrepancy, attach the note to your exam and return the note and exam to me and I will review this together with the grader. This procedure must be followed within one class period of the time the graded exams are initially returned otherwise we will not re-grade your test. In addition, we will review the entire exam to look for any other discrepancies (positive or negative).

GRADING:

We will strive to fairly grade all of your quizzes and exams. Since you will have had the problems/templates for the recitation quizzes we expect nearly perfect work ... if it is clear that you have little clue about how to work the

problem you will not receive any partial credit for that problem. The grades will range from A – F (including + and -).

The grades will be curved and the typical range for the grades are the following (G: score; $\bar{x}$: average; σ : SD):

$G > \bar{x} + \sigma$: A (typically around 90)
 $\bar{x} < G < \bar{x} + \sigma$: B, B⁺, A⁻ (typically around 75-90)
 $\bar{x} - \sigma < G < \bar{x}$: C⁻, C, C⁺, B⁻ (typically around 65 – 75)
 $G < \bar{x} - \sigma$: D ... F (typically below 65)

EVALUATION:

Recitation Quizzes	25%	- Drop two lowest rec. quiz grades.*
Lecture Quizzes	5%	- Drop two lowest lec. quiz grades.*

Midterm Exams	16.66% each (3 exams for a total of 50%) [#]
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Final Exam	<u>20%</u>
	100% - Possible Final Percentage

*If you must miss a recitation quiz then that will be the quiz grade that you will drop. Additional missed quizzes will be scored zero.

&Missed lecture quizzes result in a grade of zero for that quiz.

[#]You may substitute the grade you earn on the Final Exam for the lowest score you earn on a Midterm Exam (provided all three Midterm Exams are attempted and your Final Exam grade is higher than at least one of your Midterm Exam grades).

The University of Texas at Austin provides, upon request, appropriate academic adjustments for qualified students with disabilities. For more information, contact the Office of the Dean of Students at 471-6259, 471-4241 TDD or the College of Engineering Director of Students with Disabilities at 471-4382.

COURSE CONTENT:

1. The first law of thermodynamics and its applications (Chapters 1&2)
2. Second law, entropy and applications (Chapter 3)
3. Equations of state (Chapter 4 and 5)
4. Single Component Phase Equilibrium (Chapter 6)
5. Mixtures (Chapter 6)
6. Multicomponent Vapor-liquid; Liquid-liquid equilibrium (Chapters 7 and 8).
7. Chemical reaction equilibrium (Chapter 9).

IMPORTANT DATES:

Notes: Dates unavailable due to travel –

1. September 9 – 12 (2 classes missed – will be making up as three 50 minute lecture classes during recitations).
2. September 18 and September 19 schedule will be switched.
3. October 16 and October 15 schedule will be switched.

Plan For Recitations

Aug. 28	Review of use of steam tables + 1 lecture class
Sept. 4	Quiz over HW#1 + 1 lecture class
Sept. 11	Quiz over HW#2 (no lectures this week)
Sept. 18	Lecture makeup for September 19
Sept. 19	Quiz over HW#3 (note date and day)
Sept. 25	Quiz over HW#4 + 1 lecture class
Oct. 2	Exam #1
Oct. 9	Quiz over HW#5
Oct. 15	Quiz over HW#6 (note time and day)
Oct. 16	Lecture makeup for October 15
Oct. 23	Quiz over HW#7
Oct. 30	Exam #2
Nov. 6	Quiz over HW#8
Nov. 13	Quiz over HW#9
Nov. 20	Quiz over HW#10
Nov. 27	Thanksgiving Holiday
Dec. 4	Exam #3

The final exam is scheduled for Thursday, December 12, 3:30 pm-5:30 pm
(<https://registrar.utexas.edu/schedules/249/finals>)